



We can create an application for an e-commerce website to analyse a huge data set and predict our future shopping requirements.



Data is a collection of information gathered through observation, measurement, or questioning. We deal with data every day. It can be in the form of numbers or words.

For example, a schoolteacher works with a large amount of data every day in the form of students' attendance sheets, report cards, etc.

Different kinds of data need to be classified, sorted, organised, interpreted and represented to become useful. Processed data is called information, and the technique of extracting relevant information from large sets of data is known as **data analysis**.

What is data analysis?

Data analysis is the process used to analyse data to improve its productivity. In data analysis, data is extracted from various sources and then organised and categorised.

The main purpose of data analysis is to find meaning in large sets of data. After this process, knowledge or information extracted can be used to make better informed decisions.

Industries and large businesses use data analysis for market research, product research, customer reviews, and sentiment analysis. It is also used to predict future ups and downs that might occur based on past experiences.

Different programming languages help us to easily analyse large sets of data.

Programming languages

- A programming language allows us to give instructions to a computer in a language that the computer understands.
- Programmers use programming languages to communicate with a computer.
- These languages allow computers to process large sets of complex information quickly and efficiently.
- There are dozens of programming languages available today.
- They are used to develop websites, build games, applications, and create animations, scientific research and more.

Python is a commonly used programming language for data analysis.

Introduction to Python

Python is a general-purpose programming language. It is used to develop both, desktop and web applications. Python is also used to develop complex scientific and numeric applications, perform data analysis and create visualisations.

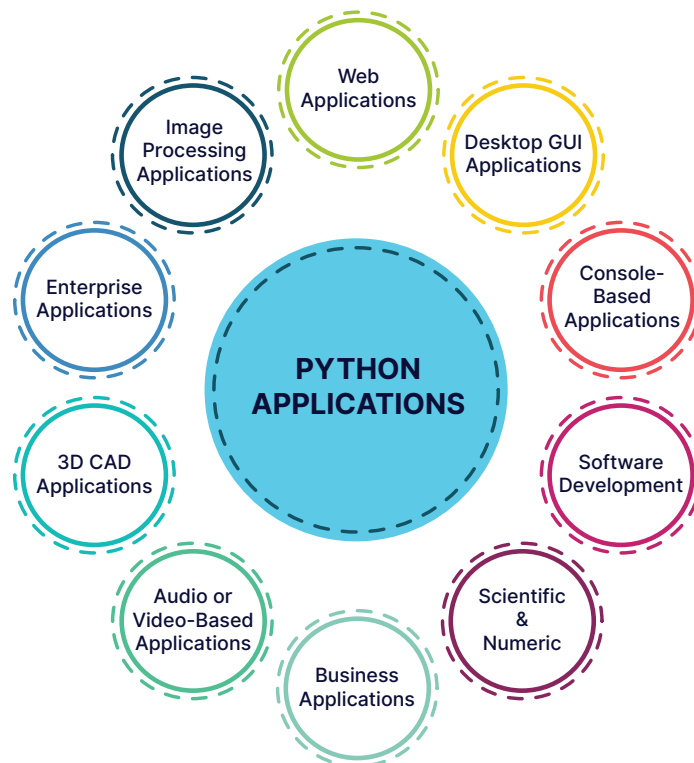


Why use Python?

- Coding in Python is easier as compared to other programming languages like Java or C++.
- Python can be used to create stunning visualisations.
- Python's syntax is easy to understand and write since it does not include curly braces and semi-colons.
- It can be installed on all major operating systems like Windows, Linux or Mac.
- Various tools are available for Python, which can be used to perform a wide variety of operations.

Applications of Python

- Python is one of the most sought-after languages for working on domains like data science, machine learning and Artificial Intelligence (AI).
- Python is used in image-processing technologies like face detection, gesture recognition.
- Python is used to develop graphical stand-alone applications.
- Python is also used to create modern web applications such as blogs, social media apps, and games.



There are two basic, but important, programming concepts we need to understand before we start working with Python:

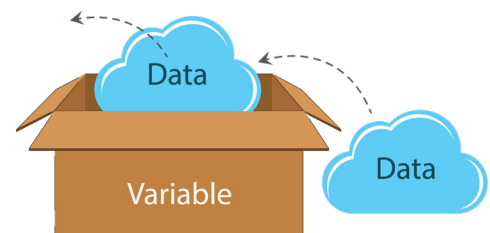
a. Variables

b. Data Types

Variables in Python

During coding, we often use a particular piece of data in different stages of a program. We change or access its value for various operations. To make this simpler, the data can be stored in a variable which can be accessed by the program at any instance.

Variables can be considered containers to store data. A variable has a name and a value that needs to be stored.



We can use an example to understand variables better:

- A variable is like a wallet.
- We use it to keep our money.
- We can access the money in the wallet at any point, and we can also change the amount in it.
- A variable is similar. We can store data in it and can access and update that data when required.



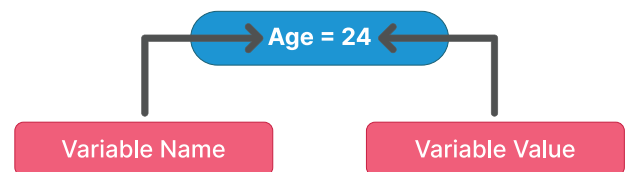
To create a variable in Python, we need to follow the following syntax: **Variable Name = Value**

Variable naming rules

- The name should be relevant to the stored value.
- It cannot start with a number or a special symbol (@, #, \$, &, etc.).
- Reserved keywords in Python cannot be used as variable names. There are 33 reserved keywords, some of which we will learn later in the lesson. These include none, import, return, and def.
- A variable name cannot have any blank spaces.
- Uppercase and lowercase variable names are different. 'NAME' and 'name' are 2 different variables.

Assigning values to a variable

To assign a value to a variable, we use the “=” symbol.



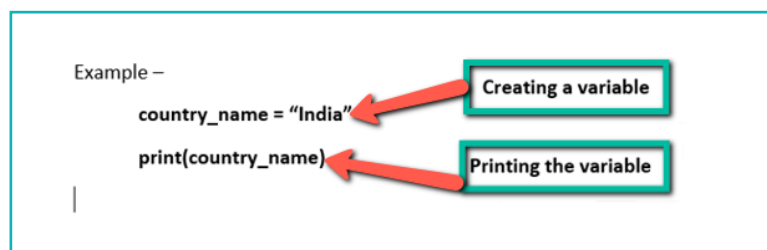
We can also store a string value, but we have to put a double (“”) or single (‘ ’) quote around the string value as shown below:

Country = “India”

Printing a variable value

Often, we will need to print or display the values stored in a variable. We use the **print()** statement to do this.

```
Example –  
country_name = "India"  
print(country_name)
```



Exercise

Tick mark ☒ the correct answer

- 1 Which of the following variable names is invalid?
- | | | | |
|----------------|--------------------------|------------|--------------------------|
| a. First_name | ☐ | b. X | ☐ |
| c. 123_address | ☐ | d. Country | ☐ |
- 2 Which of the following is the correct way to print a variable?
- | | | | |
|---------------------------|--------------------------|---------------------------|--------------------------|
| a. Print("variable_name") | ☐ | b. Print('variable_name') | ☐ |
| c. Print(variable_name) | ☐ | d. print(variable_name) | ☐ |

Data types in Python

- A data type is a classification or group of data elements. It denotes the type of value that suggests what operations can be performed on a particular piece of data.
- Variables can store different types of data.
- For example, a person's age is stored as a numeric value (integers) and their address is stored as an alphanumeric character (string).
- Python has standard data types that are used to define the operations possible on them.
- Python supports the following data types:

1. **Number:** Number data type stores two types of numeric values: integers and floats. Integers are whole numbers and floats are numbers with decimal points.

For example:

- | | |
|-------------------------------------|--|
| a. age = 10 (storing integer value) | b. weight = 50.5 (storing float value) |
|-------------------------------------|--|

2. **Strings:** Strings are identified as an adjoining set of characters, enclosed within double (") or single (') quotes.

For example:

- | | |
|--------------------|---|
| a. Name = "Python" | b. Address = "221B Baker Street, London, England" |
|--------------------|---|

3. **Boolean:** Boolean can store only 2 values, **True** and **False**

For example:

- | | |
|------------------|---------------------|
| a. IsMale = True | b. IsFemale = False |
|------------------|---------------------|

Exercise

State whether True or False

- 3 A = 12.6 is a variable with integer data type.

- 4 123_Name = Python is the correct way to assign a string variable.

- 5 Variables can be considered containers where we can store our data.

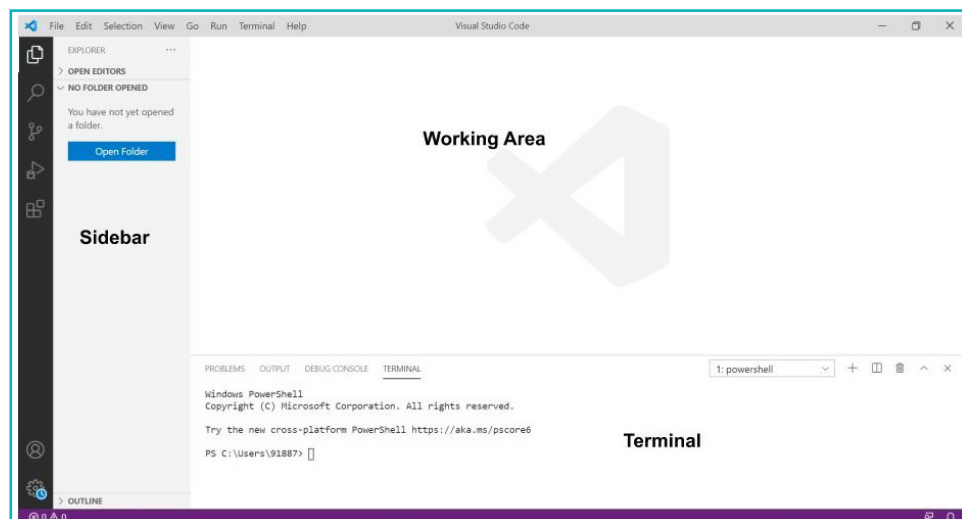
To start programming with Python, we must write and run code in an application that can read and run the Python language. An application like this is called a **code editor**.

What is a code editor?

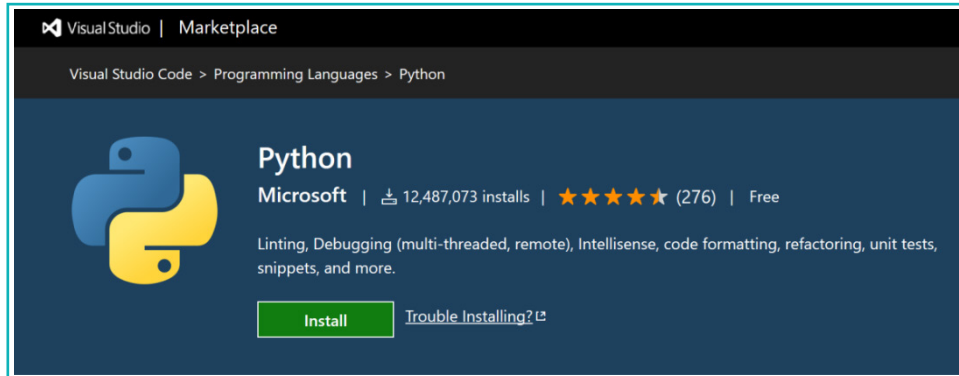
A code editor is essentially a text editor, but with lots of additional features to help us write programs. There are several code editors available for Python. We will use VS Code.

About VS Code

VS Code is a lightweight but powerful code editor used to write programs. It is freely available and supports multiple programming languages, including Python.



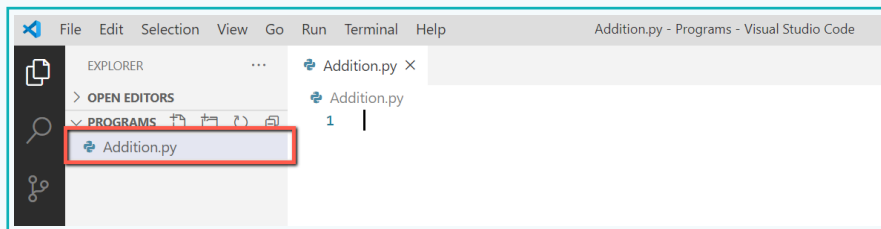
We can install the Python extension for VS Code from the Visual Studio Marketplace.



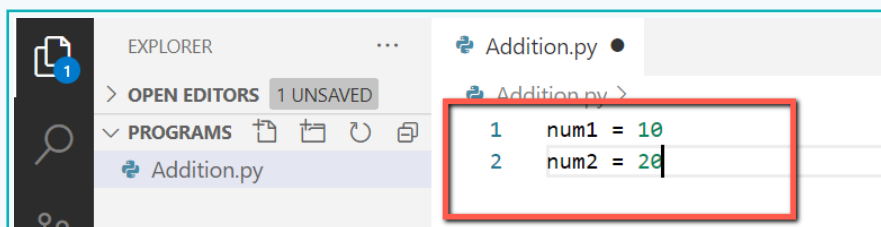
Activity 1

Create a program to add two numbers and display the result.

- Open the VS Code application and create a new file by clicking on 'File' menu > 'New File'.
- Click on File > Save to save the file.
- Name the file 'Addition' and set the extension as .py. This indicates that it's a Python program. This will create a file "Addition.py", visible in the VS Code sidebar.

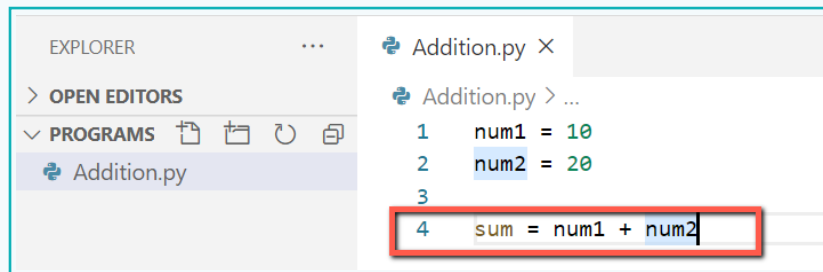


- Create two variables, num1 and num2, and store values 10 and 20 in them respectively. You may also store two different values.



Activity 1

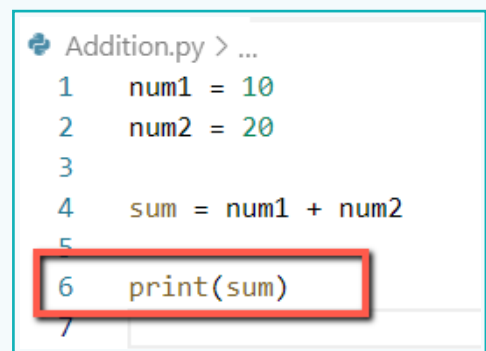
- e. To add the values stored in the two variables, use the + (addition) operator.
- f. Store the result of the operation in another variable - **sum**.



```
EXPLORER
> OPEN EDITORS
PROGRAMS
  Addition.py

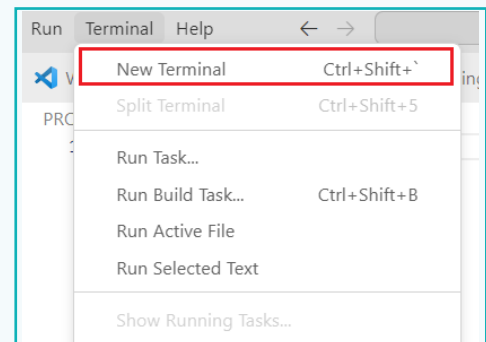
Addition.py
1 num1 = 10
2 num2 = 20
3
4 sum = num1 + num2
```

- g. Use the **print** statement to display the output (stored in the sum variable).



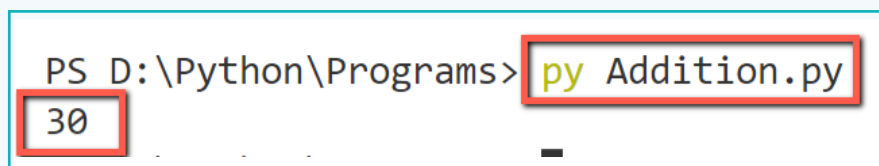
```
Addition.py
1 num1 = 10
2 num2 = 20
3
4 sum = num1 + num2
5
6 print(sum)
7
```

- h. To execute the program, we will use the inbuilt terminal in VS Code. We can access this by clicking on **Terminal > New Terminal**.



- i. To execute the program, type **py Addition.py** in the terminal and press **Enter**.
- j. The output can now be viewed in the terminal window.

Output:

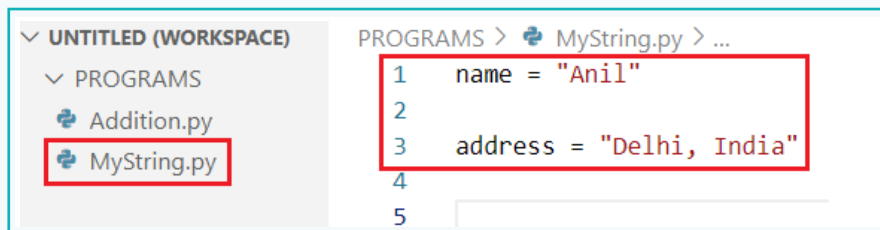


```
PS D:\Python\Programs> py Addition.py
30
```


Activity 2

Create a program to display a string of your name and address using variables.

- Create a new file and save it as 'MyString.py'.
- Open the MyString.py file in VS Code and create two variables, name and address.
- Assign your own name and address to the variables.



The screenshot shows the VS Code interface. On the left, the Explorer sidebar shows a workspace with a folder named 'PROGRAMS' containing two files: 'Addition.py' and 'MyString.py'. 'MyString.py' is selected. The main editor window shows the code for 'MyString.py' with line numbers 1 through 5. Lines 1 and 2 are highlighted with a red box: `1 name = "Anil"` and `2`. Lines 3 and 4 are highlighted with a red box: `3 address = "Delhi, India"` and `4`. Line 5 is empty.

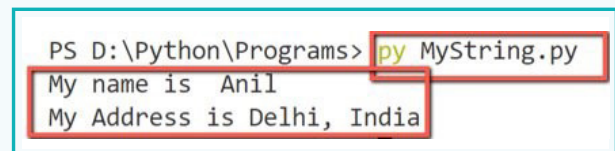
- Use the print() statement to display or print the values of the two strings.



The screenshot shows the VS Code editor with the code for 'MyString.py'. Lines 5 and 6 are highlighted with a red box: `5 print("My name is ", name)` and `6`. Lines 7 and 8 are highlighted with a red box: `7 print("My address is ", address)` and `8`.

Output:

- Open terminal in VS Code and execute the program using the command 'py MyString.py'.
- We will see the following output:



The screenshot shows a Windows command prompt window. The command prompt shows the command `PS D:\Python\Programs> py MyString.py` and the output `My name is Anil` and `My Address is Delhi, India`. The command and the first line of output are highlighted with a red box.

Activity 3

Create a program to convert temperature data from Celsius to Fahrenheit.

Let's accept input from the user

- Open VS Code, create a new file and save it as 'TempConversion.py'.
- Declare a variable 'C'. The C variable is where the user will provide the temperature in Celsius.

Activity 3

c. To take input from the user, we will use the 'input()' function.

```
TempConversion.py •
TempConversion.py > ...
1 C = input("Enter the Temperature in Celsius:")
2
```

d. Temperature value can be an integer or a float number.

e. The input from the user must be a float data type that will handle integers and float numbers.

f. Wrap the entire input() statement within another function 'float()'. Now, the value provided by the user will be stored as float data type.

Let's convert the Celsius value to Fahrenheit.

g. In the code below, first we multiply the Celsius value with 1.8 and then add 32 to it.

h. The output of this operation is stored in another variable 'F'.

```
TempConversion.py > ...
1 C = float(input("Enter the Temperature in Celsius:"))
2
3 F = (C * 1.8) + 32
4
```

i. Finally, we will display the result using the 'print()' statement.

```
3 F = (C * 1.8) + 32
4
5 print("The temperature in Farenheit is: ", F)
6
```

Output:

j. Execute the program by using the command 'py TempConversion.py' and we will be able to view the converted value as the output.

```
PS D:\Python\Programs> py TempConversion.py
Enter the temperature in Celsius 90
The temperature in Fahrenheit is 194.0
```

13. Data Analysis

In this lesson, we learnt that we are surrounded by data in every walk of life. To effectively use data, we need to categorise and organise it. We can use data analysis for this. Python is a computer language that is commonly used to categorise large sets of data. It can make our lives easy and efficient.

Data analysis helps in organising, interpreting, and presenting the data in a meaningful and simplified manner.



Tech Flash

- **Data Analysis** : The technique of extracting relevant information from large sets of data.
- **Python** : Python is a general-purpose programming language used to develop complex scientific and numeric applications, perform data analysis and create visualisations.
- **Variables** : Variables can be considered containers to store data.
- **Data Type** : Data types are the classifications or groups of data elements that denote the type of value.